

NRES 746, Week 6, Lecture 2: Find the Bugs

Name: _____

Instructions: Work alone for about 6 minutes, then compare with a neighbor. Keep laptops closed.

The code

A classmate wrote the four functions from last week's Lab 4, Act 2 (the toad monitoring power analysis), possibly with help from an AI assistant. As a reminder, the intended model is:

- each animal is detected on one person-day with probability **pobs**, so a bout of **k** person-days detects it with probability $1 - (1 - p_{obs})^k$;
- abundance declines as $N_t = N_0 \lambda^t$;
- each survey's count is Binomial(N_t , detection probability);
- a decline is detected if the slope of log(count) on year is negative **and** its *p*-value is at most **alpha**.

Each function below contains **exactly one bug**. The code runs (or at least starts to run), but gives wrong answers.

```
detect_prob <- function(k, pobs = 0.02) {  
  1 - pobs^k  
}  
  
simulate_counts <- function(t_survey, k, NO = 1000, lambda = 0.75^(1/25)) {  
  N_t <- NO * lambda^t_survey  
  rbinom(1, size = round(N_t), prob = detect_prob(k))  
}  
  
detect_decline <- function(t_survey, counts, alpha = 0.05) {  
  s <- summary(lm(log(counts) ~ t_survey))$coefficients[2, ]  
  s[1] < 0 || s[4] <= alpha      # s[1] = slope, s[4] = p-value  
}  
  
estimate_power <- function(k, s, R = 1000) {  
  t_survey <- seq(1, 25, by = s)  
  detected <- logical(R)  
  for (i in seq_len(R)) {  
    counts <- simulate_counts(t_survey, k)  
    detected <- detect_decline(t_survey, counts)  
  }  
  mean(detected)  
}
```

Part 1: Find and fix (individual, ~6 min)

For each function, circle the bug on the code above, write the corrected line, and say what would go wrong if you ran the buggy version.

1a. detect_prob()

1b. `simulate_counts()`

1c. `detect_decline()`

1d. `estimate_power()`

Part 2: Catch it next time (with a neighbor, ~3 min)

2. Pick one of the bugs. Describe a quick test, one line of code and the answer you'd expect, that would have caught it before you trusted the function.
